Distance from a point to a line

1 $d=\dfrac{|Ax_1+By_1+C|}{\sqrt{A^2+B^2}}$

2

a distance: $\dfrac{5}{\sqrt{65}}$ units

b distance: $\dfrac{62\sqrt{5}}{15}$ units

c distance: $\dfrac{28}{5}$ units

d distance: $\dfrac{72}{25}$ units

e distance: $\dfrac{23}{13}$ units

f distance: $2\sqrt{13}$ units

3 $\dfrac{9}{\sqrt{113}}$

4

a $k=7, k=-27$

b $k=5, k=35$

5

a distance: $\sqrt{5}$ units

b $b=\pm\sqrt{319}$

Geometric applications

6

a Center: $(0,0)$

b $r=0.9$

c $\dfrac{8}{\sqrt{65}}$

d 0, because the perpendicular distance from the centre of the circle to the line is bigger than the radius of the circle.

7

a Center: $(-3,-2)$

b $r=2$

c $\dfrac{6}{\sqrt{34}}$

d 2, because the perpendicular distance from the centre of the circle to the line is smaller than the radius of the circle.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8

a 1

b $x - y - 1 = 0$

c $\left(-\frac{9}{2}, -\frac{11}{2}\right)$

d -1

e Yes

f isosceles

g $BD = \frac{9}{2}\sqrt{2}$

h $AC = 3\sqrt{2}$

i Area: $\frac{27}{2}$

j $(-9, -1)$

9

a $3x - 4y + 13 = 0$

b Height: 3 units

c Area: 15 units^2

© ClipSAT: By Mr. Mohamed Abdallah +966597688647